

Emotion Classification Using a Bidirectional LSTM

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1 Task

We build a supervised text classifier for emotion detection. Given a short English sentence, the model predicts one of six labels: anger, fear, joy, love, sadness, or surprise. The submission includes preprocessing, feature analysis, model training, and evaluation using multiple metrics and visualisations.

2 Dataset and Split

We use the *Emotion Dataset for NLP* (six emotion classes). Data is split using stratified sampling into train/validation/test (80/10/10). The validation set is used for early stopping; the test set is held out for final reporting.

3 Preprocessing and Feature Analysis

Preprocessing includes: (i) lowercasing, URL/punctuation/non-alphabetic removal, (ii) NLTK tokenisation, (iii) stop-word removal and lemmatisation. Figure 1 shows token length distribution after preprocessing; most texts are short, motivating truncation at max length 50.

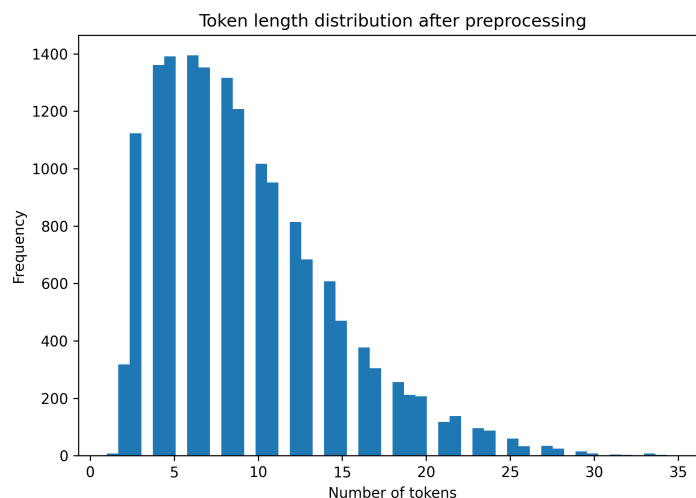


Figure 1: Token length distribution after preprocessing.

Feature analysis indicates label imbalance and lexical overlap between semantically close emotions (e.g. joy/love), motivating contextual sequence modelling and use of macro/weighted metrics.

4 Model and Training

We implement a 2-layer BiLSTM classifier: learned 100-d embeddings, hidden size 128, dropout 0.5, bidirectional output concatenation, and a final linear layer for 6-way prediction. Training uses cross-entropy loss and Adam ($lr = 10^{-3}$), gradient clipping, and early stopping based on validation accuracy. Vocabulary is built from the training split only to avoid leakage.

5 Evaluation and Results

We report accuracy, micro/macro/weighted F1, balanced accuracy, and Matthews correlation coefficient (MCC). Figures 2 and 3 show smooth convergence.

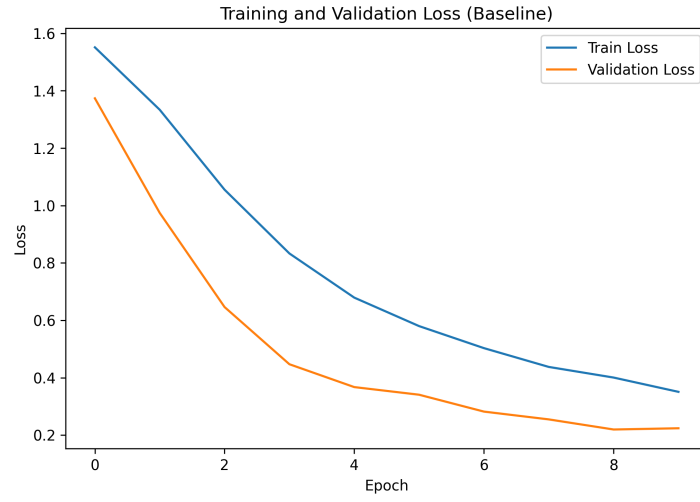


Figure 2: Training and validation loss.

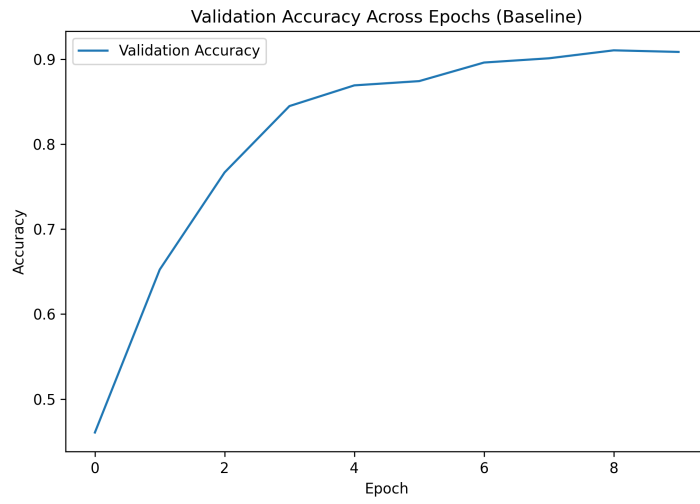


Figure 3: Validation accuracy across epochs.

Metric	Value
Accuracy	0.9038
F1 (Micro)	0.9038
F1 (Macro)	0.8686
F1 (Weighted)	0.9042
Balanced Accuracy	0.8739
MCC	0.8738

Table 1: Final test-set performance metrics.

Figure 4 shows the confusion matrix; most confusions occur between semantically similar emotions (joy vs. love; fear vs. sadness), consistent with limited context and label overlap.

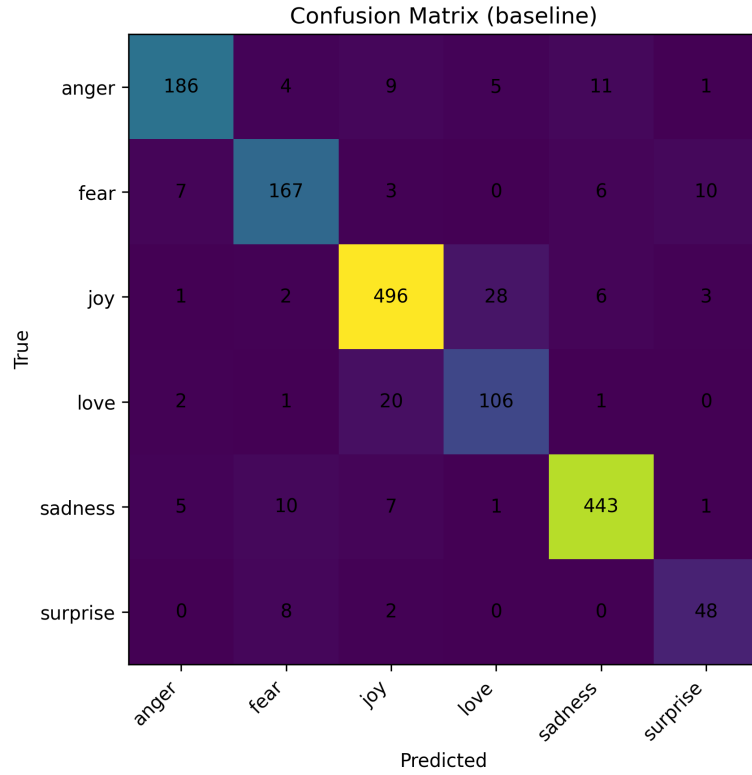


Figure 4: Confusion matrix on the test set.

6 Conclusion

The BiLSTM achieves strong and balanced performance (macro F1 0.8686, balanced accuracy 0.8739). Errors primarily reflect semantic overlap between emotions and short text context. Future work could explore pretrained embeddings or transformer encoders and qualitative analysis of ambiguous cases.